

On computing and the complexity of computing higher-order U -statistics, exactly

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Paper and resources

1 What are U -statistics?

A. Definition

$$\mathbb{U}_{n,m}[h] = \sum_{\substack{(i_1, i_2, \dots, i_m) \in [n]^m \\ \text{all indices distinct}}} h(X_{i_1}, \dots, X_{i_m})$$

Here $h : \mathcal{X}^m \rightarrow \mathbb{R}$ is a kernel function and $[n] = \{1, \dots, n\}$. We omit the normalization constant $(n)_m = n!/(n-m)!$ in the computational notation. If $X_1, \dots, X_n$ are i.i.d. and $\mathbb{E}[h(X_1, \dots, X_m)] < \infty$, then

$$\underbrace{\frac{1}{(n)_m} \mathbb{E}[\mathbb{U}_{n,m}[h]]}_{\text{unbiasedness}} = \mathbb{E}[h(X_1, \dots, X_m)].$$

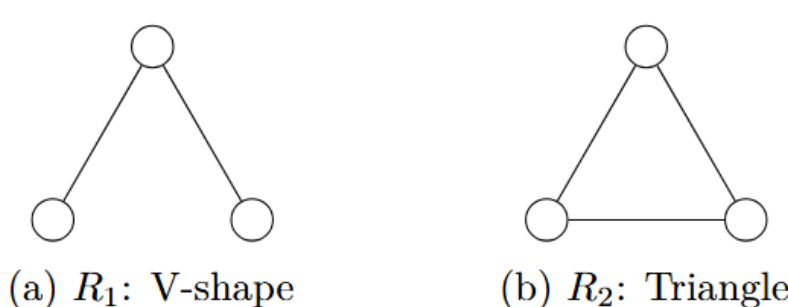
B. Examples

(a) Higher-order influence functions (HOIFs) [1].

$$h_m^{\text{HOIF}}(X_1, \dots, X_m) = [a_1 \phi(Z_1)^\top \phi(Z_2)] [\phi(Z_2)^\top \phi(Z_3)] \cdots [\phi(Z_{m-1})^\top \phi(Z_m) b_m] \\ = f_1(X_1, X_2) \cdots f_{m-1}(X_{m-1}, X_m).$$

Here a_i, b_i are scalars, and ϕ is vector-valued.

(b) Motif counting: V-shapes and triangles. For adjacency matrix A of a simple undirected graph and $B_{ij} = 1 - A_{ij}$,



$$C(R_1) = \frac{1}{2} \sum_{\substack{(i_1, i_2, i_3) \in [n]^3 \\ \text{all indices distinct}}} A_{i_1 i_2} A_{i_2 i_3} B_{i_3 i_1}, \\ C(R_2) = \frac{1}{6} \sum_{\substack{(i_1, i_2, i_3) \in [n]^3 \\ \text{all indices distinct}}} A_{i_1 i_2} A_{i_2 i_3} A_{i_3 i_1}.$$

(c) Squared distance covariance [2].

$$\text{dCov}^2(X, Y) = \frac{(n-4)!}{n!} \sum_{i,j,q,r \text{ distinct}} (a_{ij} b_{qr} + a_{ij} b_{ij} - a_{ij} b_{iq} - a_{ij} b_{jr}),$$

Here $a_{ij} = \|X_i - X_j\|_2$ and $b_{ij} = \|Y_i - Y_j\|_2$, where $(X_1, Y_1), \dots, (X_n, Y_n)$ are i.i.d. observations of random vectors (X, Y) .

2 How to compute U -statistics, exactly

A. Incomplete U -statistics: sample fewer tuples

$$\frac{1}{|S|} \sum_{(i_1, \dots, i_m) \in S} h(X_{i_1}, \dots, X_{i_m}), \quad |S| \ll (n)_m.$$

Drawbacks:

- (a) Fewer sampled tuples use less information and generally increase variance.
- (b) Randomization requires additional statistical analysis.
- (c) Direct enumeration remains inefficient; a general computational tool is still needed.

B. A close relative: V -statistics and einsum

$$\mathbb{V}_{n,m}[h] = \sum_{(i_1, \dots, i_m) \in [n]^m} h(X_{i_1}, \dots, X_{i_m}).$$

For $h_4^{\text{HOIF}} = f_1 f_2 f_3$, let $T_{ij}^{(k)} = f_k(X_i, X_j)$, $k = 1, 2, 3$. Then

$$\mathbb{V}_{n,4}[h_4^{\text{HOIF}}] = \sum_{i_1, i_2, i_3, i_4} T_{i_1 i_2}^{(1)} T_{i_2 i_3}^{(2)} T_{i_3 i_4}^{(3)}.$$

`einsum('ij,jk,kl->', T1, T2, T3)`

`numpy.einsum` and `torch.einsum` implement tensor contractions; PyTorch supports parallel CPU and GPU computation.

C. Exact U -to- V conversion: Möbius inversion

General formula

Π_m is the set of partitions of $[m]$, and $\mathbb{V}[\pi](h)$ sums over indices equal within each block of π , with no constraints between blocks.

$$\mathbb{U}[h] = \sum_{\pi \in \Pi_m} \mu_\pi \mathbb{V}[\pi](h), \quad \mu_\pi = (-1)^{m-|\pi|} \prod_{C \in \pi} (|C| - 1)!.$$

4-th order HOIF and sparsification

Set the diagonal entries of the factor tensors to zero without changing the distinct-index sum:

$$\tilde{T}_{ij}^{(k)} = T_{ij}^{(k)} \mathbf{1}\{i \neq j\}, \quad k = 1, 2, 3.$$

For example, when $i_1 = i_2$, the first factor is zero:

$$\mathbb{V}[\{1, 2\}|\{3\}|\{4\}](\tilde{h}_4^{\text{HOIF}}) = \sum_{i,j,k} \underbrace{\tilde{T}_{ij}^{(1)}}_0 \tilde{T}_{ij}^{(2)} \tilde{T}_{jk}^{(3)} = \sum_{i,j,k} 0 \cdot \tilde{T}_{ij}^{(2)} \tilde{T}_{jk}^{(3)} = 0.$$

The full expansion cancels ten such terms:

$$\mathbb{U}[h_4^{\text{HOIF}}] = \left(\sum_{i_1, i_2, i_3, i_4} - \sum_{i_1 \neq i_2, i_3, i_4} - \sum_{i_1 = i_3, i_2, i_4} - \sum_{i_1 = i_4, i_2, i_3} - \sum_{i_2 \neq i_3, i_1, i_4} - \sum_{i_2 = i_4, i_1, i_3} - \sum_{i_3 \neq i_4, i_1, i_2} - \sum_{i_3 = i_4, i_1, i_2} \right. \\ \left. + \sum_{i_1 \neq i_2, i_3 = i_4} + \sum_{i_1 = i_3, i_2 = i_4} + \sum_{i_1 \neq i_4, i_2 = i_3} + 2 \sum_{i_1 \neq i_2 = i_3, i_4} + 2 \sum_{i_1 \neq i_2 = i_4, i_3} \right. \\ \left. + 2 \sum_{i_1 \neq i_3 = i_4, i_2} + 2 \sum_{i_1 \neq i_3 = i_4, i_1} - 6 \sum_{i_1 \neq i_2 = i_3 = i_4} \right) \cdot \tilde{T}_{i_1 i_2}^{(1)} \tilde{T}_{i_2 i_3}^{(2)} \tilde{T}_{i_3 i_4}^{(3)}.$$

Decomposition signature and final formula

We need a mathematical object to characterize the multiplicative decomposable structure of the kernel. For example, for the 4-th order HOIF, define the decomposition signature as

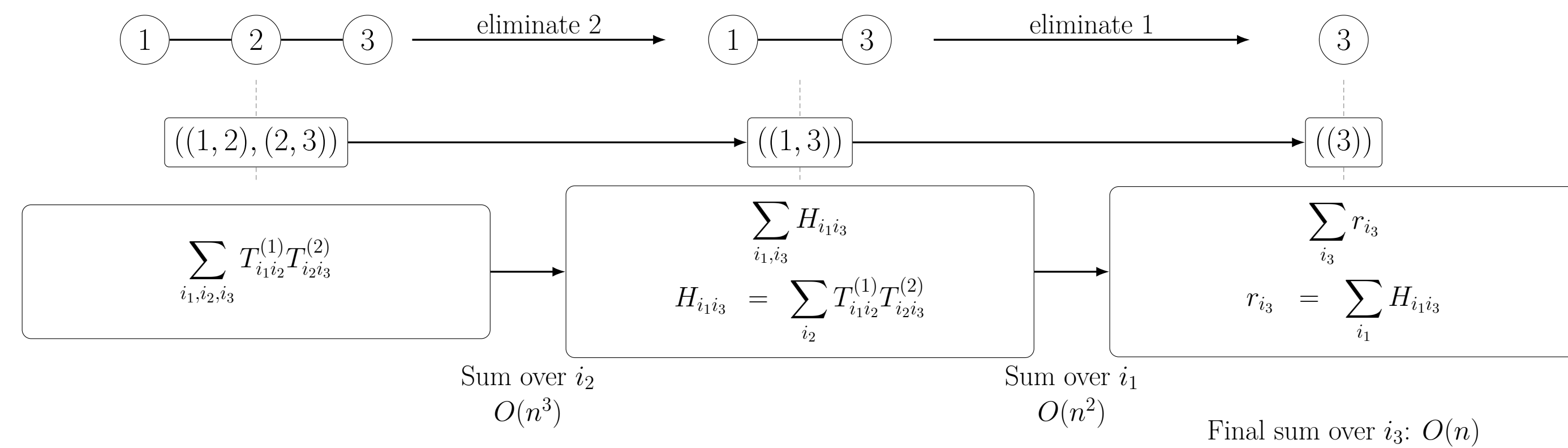
$$\mathcal{A}_4^{\text{HOIF}} = ((1, 2), (2, 3), (3, 4)).$$

For a kernel with decomposition signature $\mathcal{A}$, let $\tilde{h}$ be its product of diagonal-excluded factors. The same Möbius coefficients apply; only the partition range changes:

$$\mathbb{U}[h] = \sum_{\pi \in \Pi_m^{\mathcal{A}}} \mu_\pi \mathbb{V}[\pi](\tilde{h}). \\ \Pi_m^{\mathcal{A}} = \{\pi \in \Pi_m : \forall Q \in \pi, \forall A \in \mathcal{A}, |Q \cap \text{set}[A]| < 2\}.$$

3 Complexity analysis

A. Complexity of V -statistics: Treewidth



For this 3-rd order V -statistic with decomposition signature $((1, 2), (2, 3))$, the ordering $(2, 1, 3)$ costs $O(n^3)$; the optimal ordering $(1, 2, 3)$ costs $O(n^2)$.

$$\text{Complexity}(\mathbb{V}) = O(n^{\text{tw}(G)+1})$$

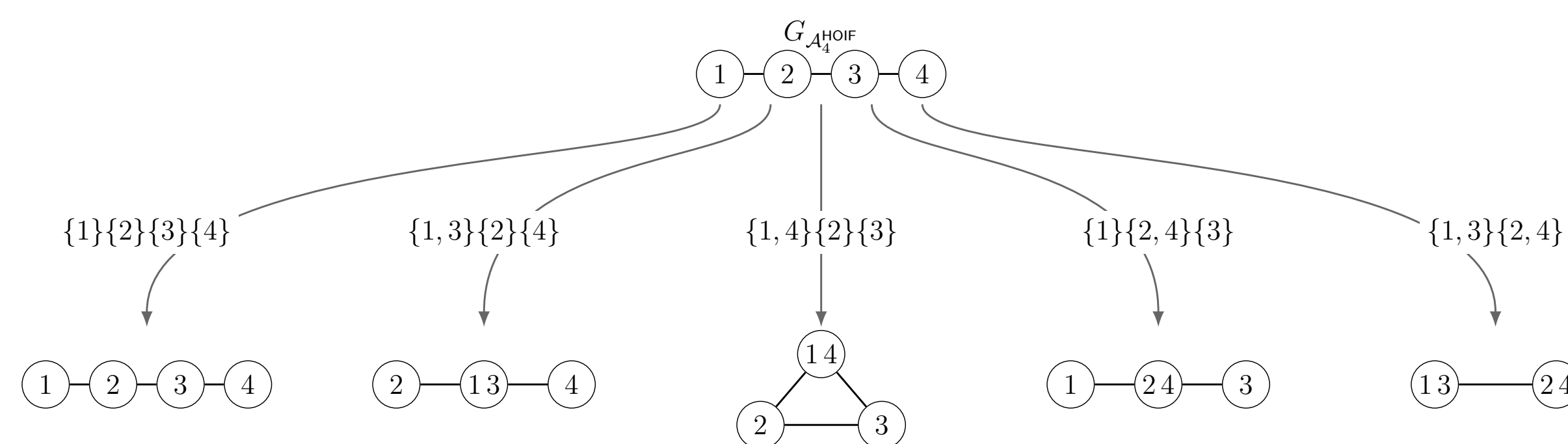
The connection between optimal computational complexity and treewidth was first discovered in [3].

B. Complexity of U -statistics: Quotient graphs

For a U -statistic with decomposition signature $\mathcal{A}$, let the corresponding graph be $G_{\mathcal{A}}$. All graphs of its induced V -statistics can be characterized by quotient graphs:

$$\text{Graph}(\mathbb{V}[\pi](h)) = G_{\mathcal{A}}/\pi.$$

For example, for the 4-th order HOIF, $\mathcal{A} = ((1, 2), (2, 3), (3, 4))$:



The triangle has treewidth 2, while the other four quotient graphs have treewidth 1. Therefore:

$$\text{Complexity}(\mathbb{U}[h_4^{\text{HOIF}}]) = O(n^3 + 4n^2).$$

C. A clean graph-theoretic question

An upper bound on the complexity of computing a general U -statistic thus reduces to a graph-theoretic problem:

Given a simple graph G and a decomposition signature $\mathcal{A}$, among the quotient graphs G/π with $\pi \in \Pi_m^{\mathcal{A}}$:

- (a) What is the maximum treewidth?
- (b) What is the distribution of treewidth?

4 Performance in practice

Python and R packages: `u-stats` (PyPI) `ustats` (CRAN)

A. HOIF: $m = 6, 7$, both $O(n^3)$

n	$m = 6$		$m = 7$	
	CPU	GPU	CPU	GPU
1000	0.453	0.019	1.733	0.073
2000	1.527	0.025	6.808	0.103
4000	7.769	0.138	36.413	0.647
8000	41.786	0.890	197.059	4.493
10000	62.540	1.665	288.897	8.589

Seconds, mean of 10 runs. CPU: Xeon 8358, 64 cores; GPU: RTX 5090, 32GB. Excludes kernel evaluation and tensorization.

B. Triangle counting: $n = 10000$, $O(n^3)$

Edge probability p	u-stats (s)	cuGraph (s)
0.001	2.8014	0.1536
0.010	3.3363	1.0379
0.100	7.2829	10.3661
0.800	37.6880	OOM

Four density settings on Erdős–Rényi graphs; RTX 4090, 24GB. cuGraph [4] is faster on sparse inputs; u-stats remains feasible at high density. OOM: out of memory.

C. Distance covariance: exact ($O(n^2)$) vs. incomplete

u-stats (exact)		MATLAB [5]			
Single	Parallel	Incomplete $\alpha = 1.5$	Incomplete $\alpha = 2.0$	Incomplete $\alpha = 2.5$	Complete
4.0928	0.1847	0.4211	4.5744	53.0265	3395.4001

$n = 138$; point estimation only; 10-run mean on Xeon 8358 (64 cores, 512GB). Incomplete methods sample $O(n^a)$ tuples [5].

References

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